

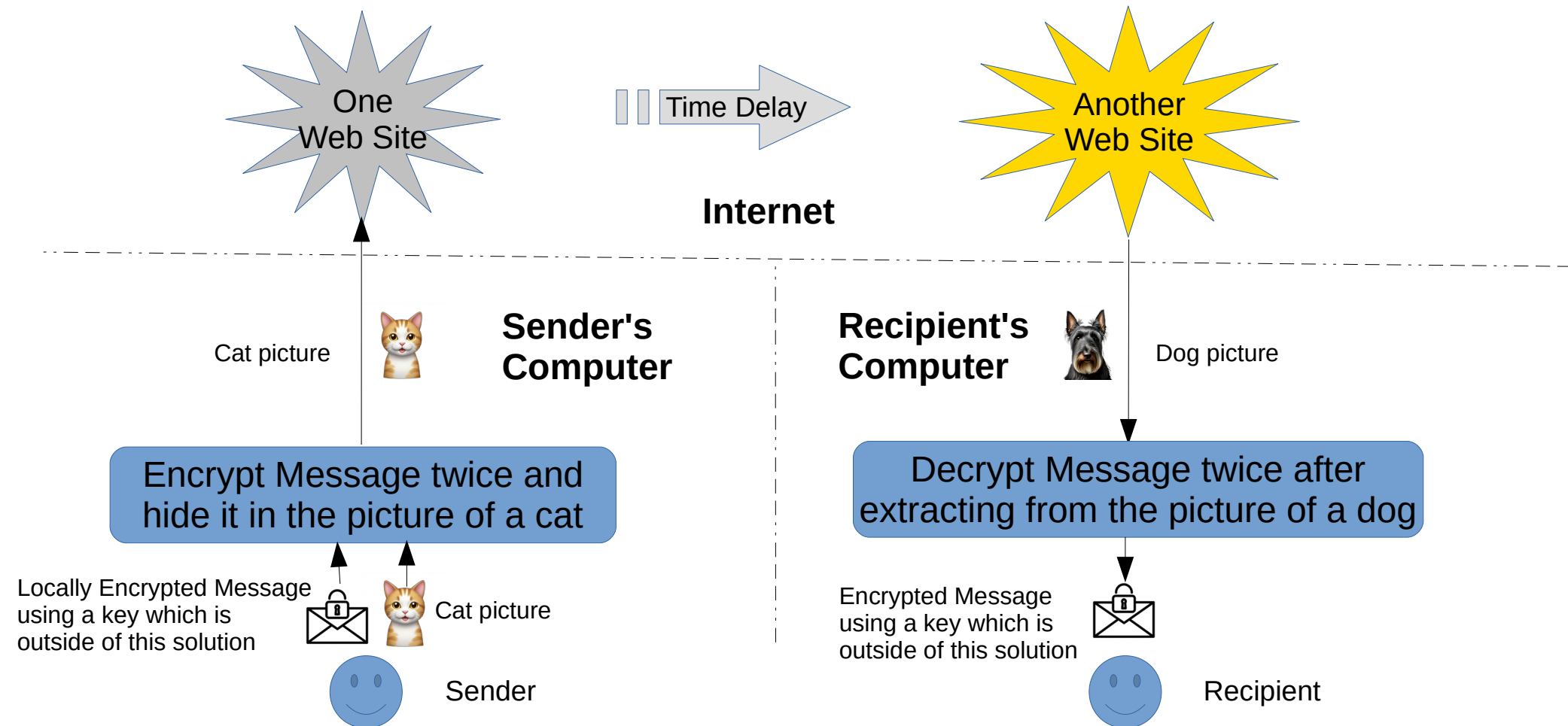
Messaging In Plain Sight (MIPS)



Objectives

1. To allow an anonymous group to communicate
2. To obfuscate that the group ever communicated
3. To keep the contents of the conversation secret
4. To obfuscate that individuals used encryption

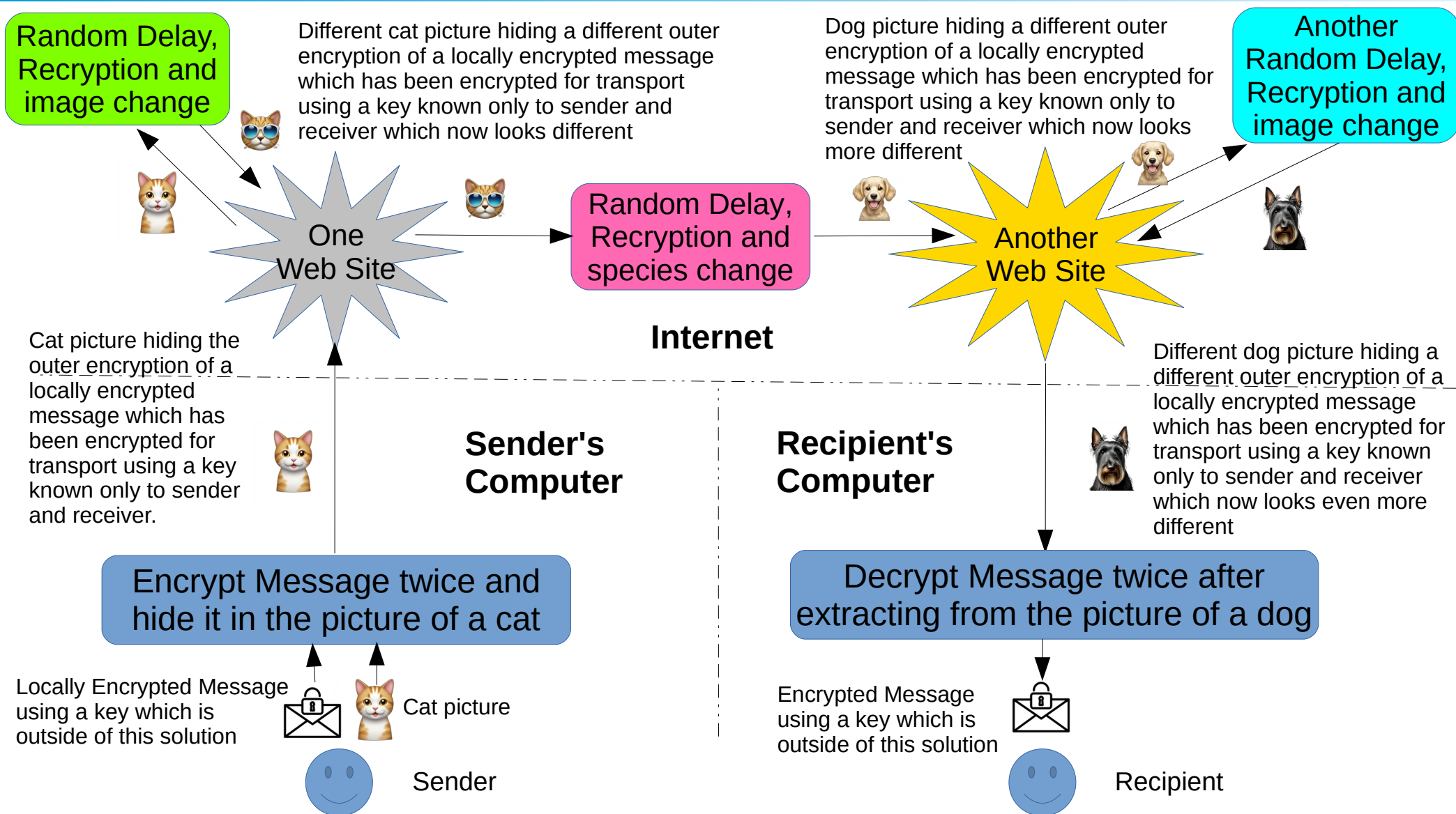
Solution



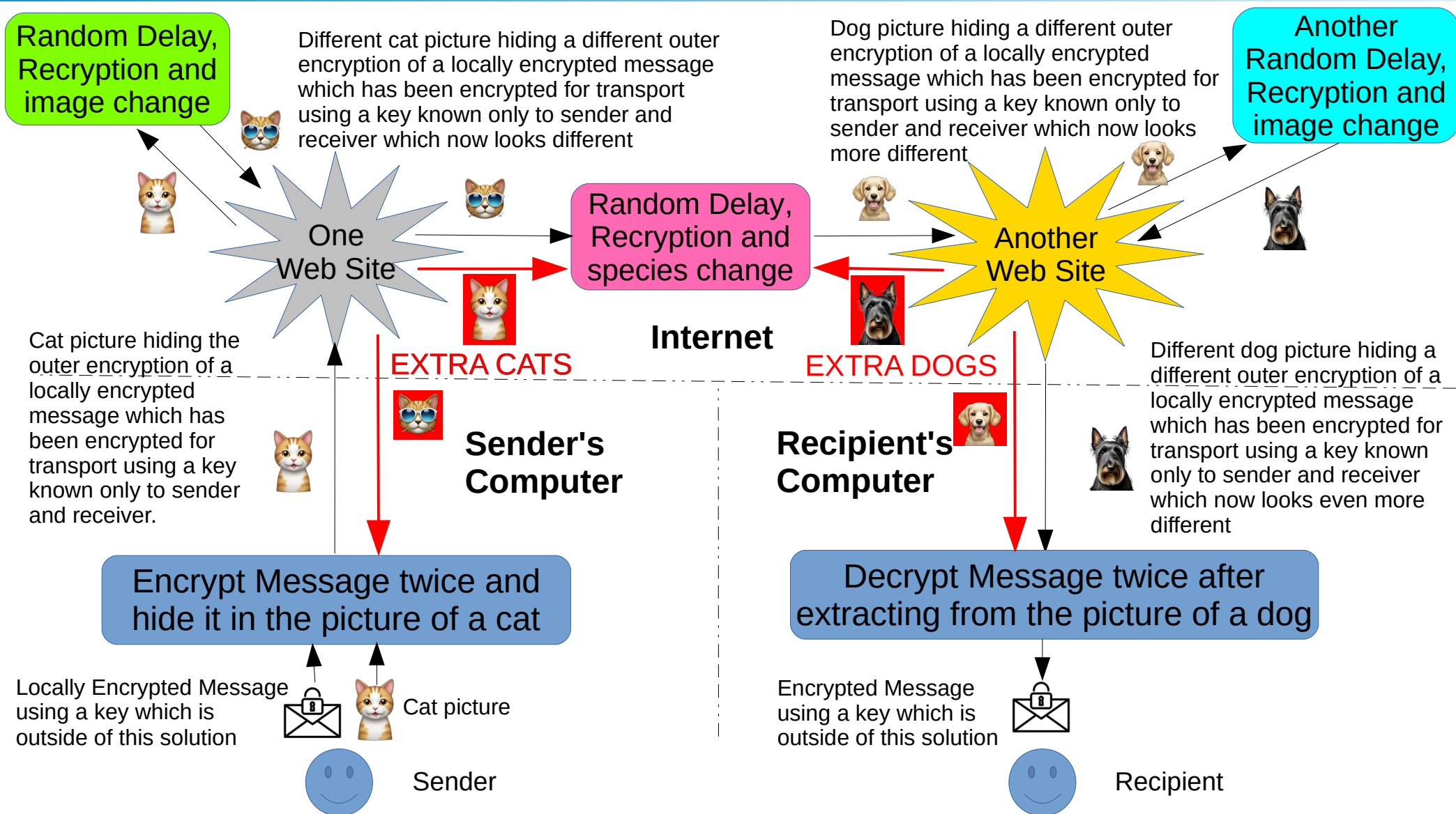
Access is Anonymous

- Users have no identity, there are no names
 - Because of this they are never authenticated
 - If they misbehave, they are disconnected
- To send, a user needs a key to a valid account
 - Accounts are issued to people or organizations
 - Account usage could be metered and billed

Default Design



Alternative – All nodes treated the same



Visible Part of Message

The IDAT Section of the outer PNG can hold the veneer of a picture



← High-order bytes of each pixel make it look like a real image



Low-order bytes of each pixel look like random noise but are actually an AES-256 encrypted PNG Image.

The key used to encrypt this inner PNG is also known to the next node in the path which will use it to decrypt the inner PNG and hide it behind a different image before passing it along to the node which follows it.

Inner Part of Message

The IDAT Section of the inner PNG can hold a pastel of cipher text



All of the bytes each pixel look like random noise but are actually an AES-256 encrypted hidden PNG which holds the padded sender-encrypted file contents as its pixel bytes.

The key used to encrypt this PNG is known only to the sender and the receiver and the client part of this solution which runs on their computers. This solution has no knowledge of the sender's encryption which should ideally be done on a different computer.

Transport Process Example

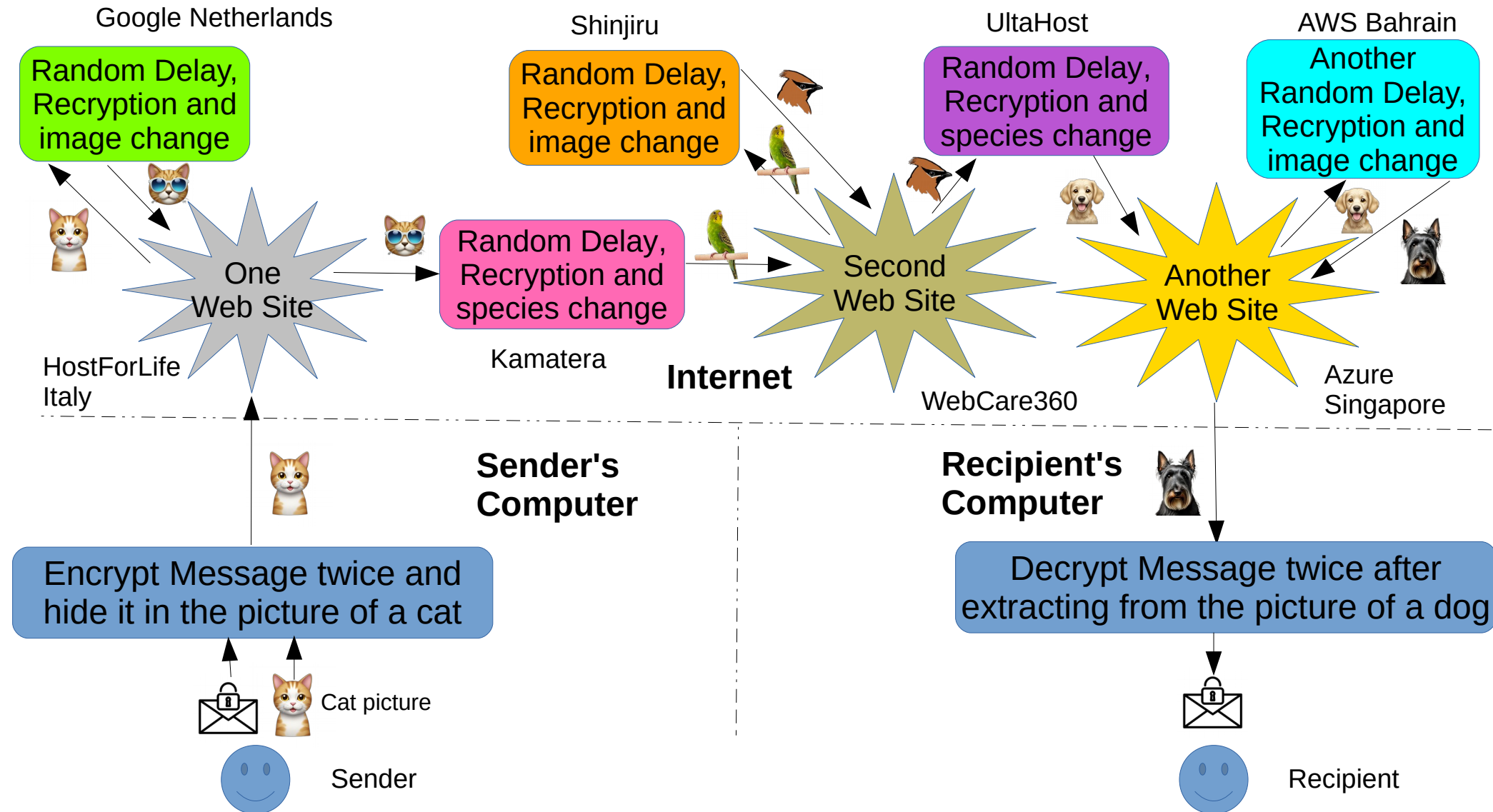
- Original Cat PNG

- Normal House Cat in high-order pixel bytes
- Inner PNG encrypted with a key known to the green delay node in low-order bytes
 - Inner PNG contains a PNG hidden in all of its pixel bytes which is encrypted with a key known to only the sender, the recipients and the client program which they run on their computers which contains a message
 - The message is the padded sender-encrypted file contents

- Transformed Cat PNG

- Cool Cat with sunglasses in high-order pixel bytes
- Inner PNG encrypted with a key known to the pink delay node in low-order bytes
 - Inner PNG contains a PNG hidden in all of its pixel bytes which is encrypted with a key known to only the sender, the recipients and the client program which they run on their computers which contains a message
 - The message is the padded sender-encrypted file contents

Wide Area Network Deployment



There are Four Types of Nodes

- 1) Light Blue Clients which send and receive
- 2) Silver, Khaki and Gold Web Sites*
- 3) Lime, Orange and Aqua image-changing Helpers
- 4) Pink and Purple species-changing Proxies

*SignalR Hubs for now

Keys Used

Message	Outer PNG	Inner PNG	Hidden PNG that holds message
	None. Looks like 	Green Delay Key*	Key just for sender and recipients
	None. Looks like 	Pink Delay Key	Key just for sender and recipients
	None. Looks like 	Orange Delay Key	Key just for sender and recipients
	None. Looks like 	Purple Delay Key	Key just for sender and recipients
	None. Looks like 	Aqua Delay Key	Key just for sender and recipients
	None. Looks like 	Mail Delivery Key	Key just for sender and recipients

*This is the key to the account issued to the user's organization whose use may be metered
There may be several (or many) valid Green Delay Keys as each organization has its own

Local Events

Sender's Computer

Time	Message	From	To
Start		Sender	Silver Web Site

Recipients Computers

Time	Message	From	To
17 seconds after start		Gold Web Site	Recipients

Network Events in Default Mode

Time	Message	From	To	Goes to
Start		Sender	Silver Web Site	Green Delay
4 seconds after start		Green Delay	Silver Web Site	Pink Delay
6 seconds after start		Pink Delay	Khaki Web Site	Orange Delay
9 seconds after start		Orange Delay	Khaki Web Site	Purple Delay
11 seconds after start		Purple Delay	Gold Web Site	Aqua Delay
17 seconds after start		Aqua Delay	Gold Web Site	Receipients

In Default mode:

- Messages are sent only to nodes in roles which can use them
- Messages are always changed before being sent downstream

In Alternative mode (next slide)

- The delay nodes are difficult to identify since there is no special routing
- All nodes waste resources discovering that messages are not for them
- Un-encrypted messages are sent downstream exposing connections
- Brute force decryption is more difficult due to the irrelevant messages

Network Events in Alternative Mode

Time	Message	From	To	Goes to
Start		Sender	Silver Web Site	Green Delay, Other Senders and Pink Delay
4 seconds after start		Green Delay	Silver Web Site	Pink Delay and All Senders
6 seconds after start		Pink Delay	Khaki Web Site	Orange Delay
9 seconds after start		Orange Delay	Khaki Web Site	Purple Delay
11 seconds after start		Purple Delay	Gold Web Site	Aqua Delay and All Recipients
17 seconds after start		Aqua Delay	Gold Web Site	Pink Delay and All Recipients

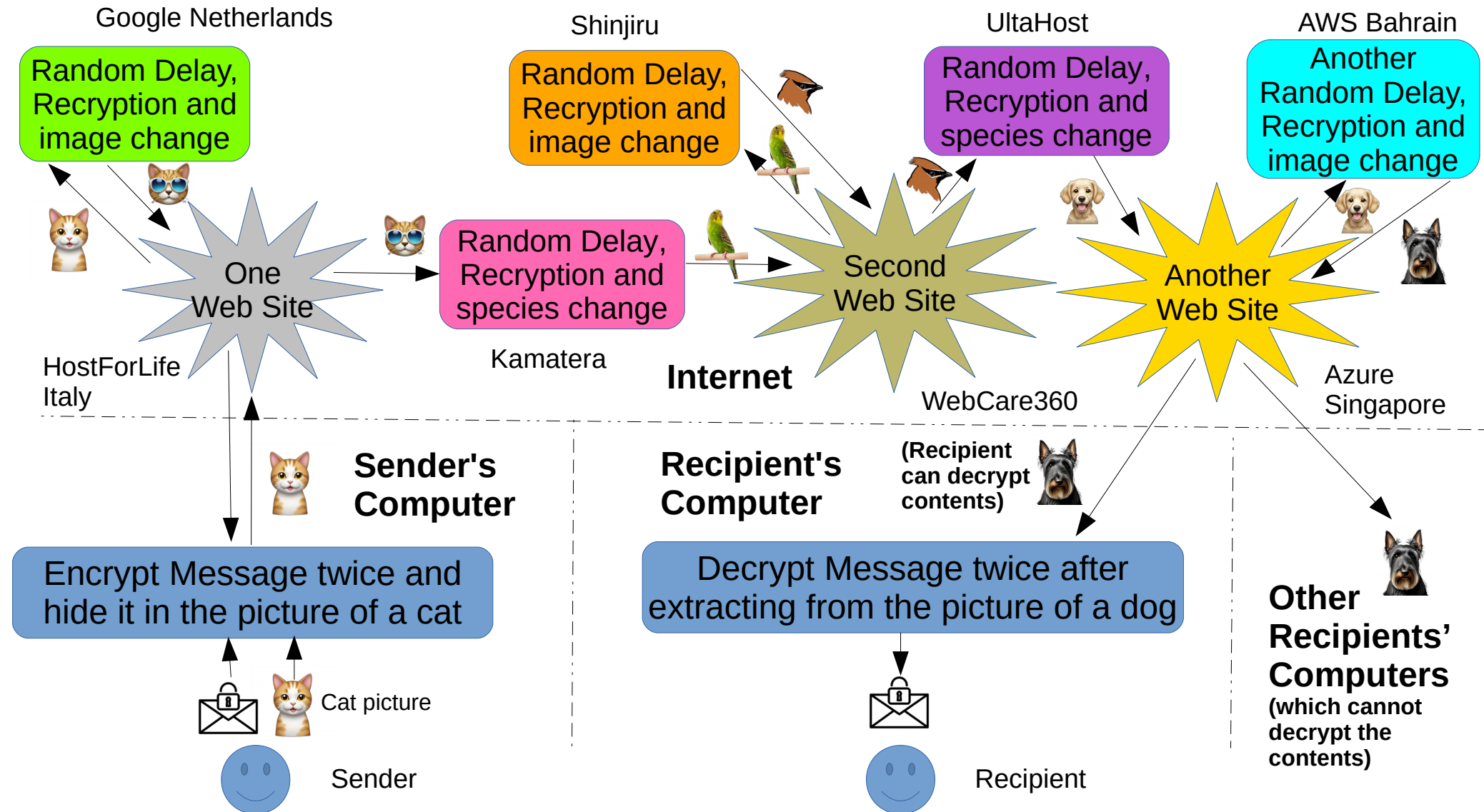
Differences Between Modes

- Default mode treats nodes connected to the web sites differently which exposes that nodes have specific roles
- Alternative mode treats all nodes connected to the web site the same
- Default mode minimizes the nodes which see a message before it is recrypted so the original message is not forwarded
- Alternative mode forwards both the original and recrypted versions of the message

And Most Importantly

- In Default mode, every recipient gets it's and all other recipient's messages
- In Alternate mode, every recipient gets two different encryptions of it's and all other recipient's messages and every sender gets the first encryption of each message sent by others and the second encryption of all messages sent
- In both cases,
 - Only the intended recipient will be able to decode their messages
 - Each recipient will need try to decrypt all of the messages to find out which are theirs
 - Their messages will be the ones which they can decrypt

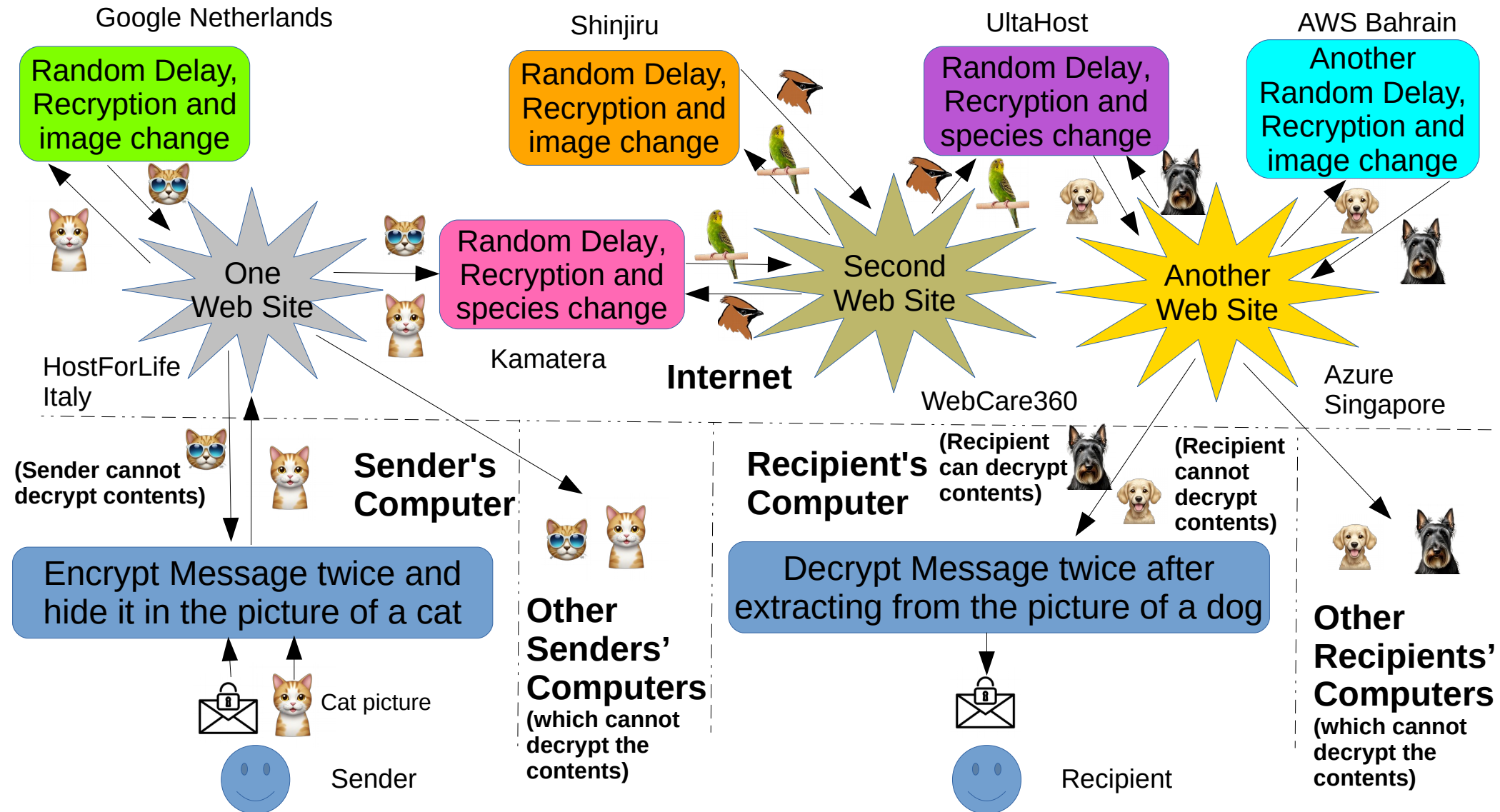
Default Mode Flow



Default Mode Chain of Custody

Time	Site	Content	Hosted by
Start	Sender		Sender
Start	Silver Web Site		HostForLife Italy
Start	Green Delay		Google Netherlands
4 seconds after start	Green Delay		Google Netherlands
4 seconds after start	Silver Web Site		HostForLife Italy
4 seconds after start	Pink Delay		Kamatera (Bulletproof)
6 seconds after start	Pink Delay		Kamatera (Bulletproof)
6 seconds after start	Khaki Web Site		WebCare360 (Bulletproof)
6 seconds after start	Orange Delay		Shinjiru (Bulletproof)
9 seconds after start	Orange Delay		Shinjiru (Bulletproof)
9 seconds after start	Khaki Web Site		WebCare360 (Bulletproof)
9 seconds after start	Purple Delay		UltraHost (Bulletproof)
11 seconds after start	Purple Delay		UltraHost (Bulletproof)
11 seconds after start	Gold Web Site		Azure Singapore
11 seconds after start	Aqua Delay		AWS Bahrain
17 seconds after start	Aqua Delay		AWS Bahrain
17 seconds after start	Gold Web Site		Azure Singapore
17 seconds after start	Recipients		All Recipients

Alternate Mode Flow



Alternate Mode Chain of Custody 1 of 2

Time	Site	Content	Hosted by
Start	Sender		Sender
Start	Silver Web Site		HostForLife Italy
Start	Green Delay		Google Netherlands
Start	Pink Delay		Kamatera (Bulletproof)
Start	Other Senders		Other Senders
4 seconds after start	Green Delay		Google Netherlands
4 seconds after start	Silver Web Site		HostForLife Italy
4 seconds after start	Pink Delay		Kamatera (Bulletproof)
4 seconds after start	Sender		Sender
4 seconds after start	Other Senders		Other Senders
6 seconds after start	Pink Delay		Kamatera (Bulletproof)
6 seconds after start	Khaki Web Site		WebCare360 (Bulletproof)
6 seconds after start	Orange Delay		Shinjiru (Bulletproof)
6 seconds after start	Purple Delay		UltraHost (Bulletproof)

Additional connections (absent in Default Mode) are shown in **bold**

Alternate Mode Chain of Custody 2 of 2

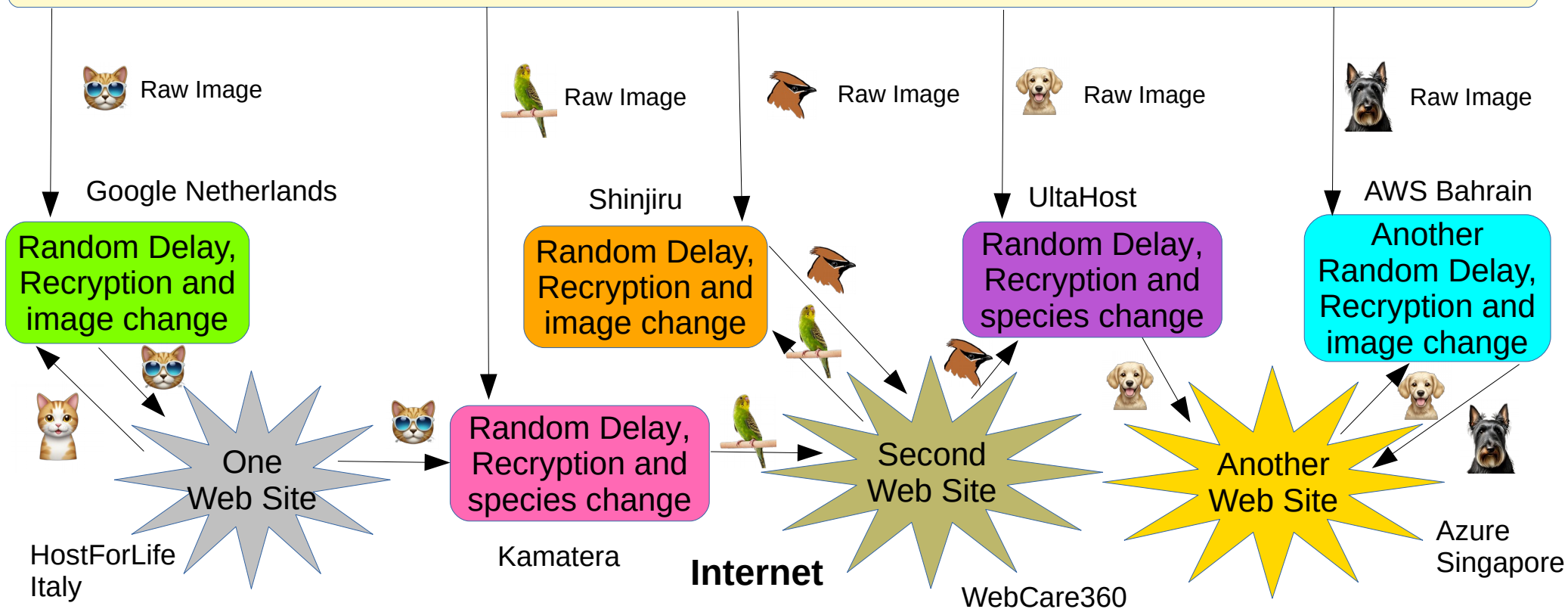
Time	Site	Content	Hosted by
9 seconds after start	Orange Delay		Shinjiru (Bulletproof)
9 seconds after start	Khaki Web Site		WebCare360 (Bulletproof)
9 seconds after start	Purple Delay		UltraHost (Bulletproof)
9 seconds after start	Pink Delay		Kamatera (Bulletproof)
11 seconds after start	Purple Delay		UltraHost (Bulletproof)
11 seconds after start	Gold Web Site		Azure Singapore
11 seconds after start	Aqua Delay		AWS Bahrain
11 seconds after start	Recipients		All Recipients
17 seconds after start	Aqua Delay		AWS Bahrain
17 seconds after start	Gold Web Site		Azure Singapore
17 seconds after start	Recipients		All Recipients
17 seconds after start	Purple Delay		UltraHost (Bulletproof)

It is possible for the carriers (together) to prove the end to end linkage between nodes

AI Images Source

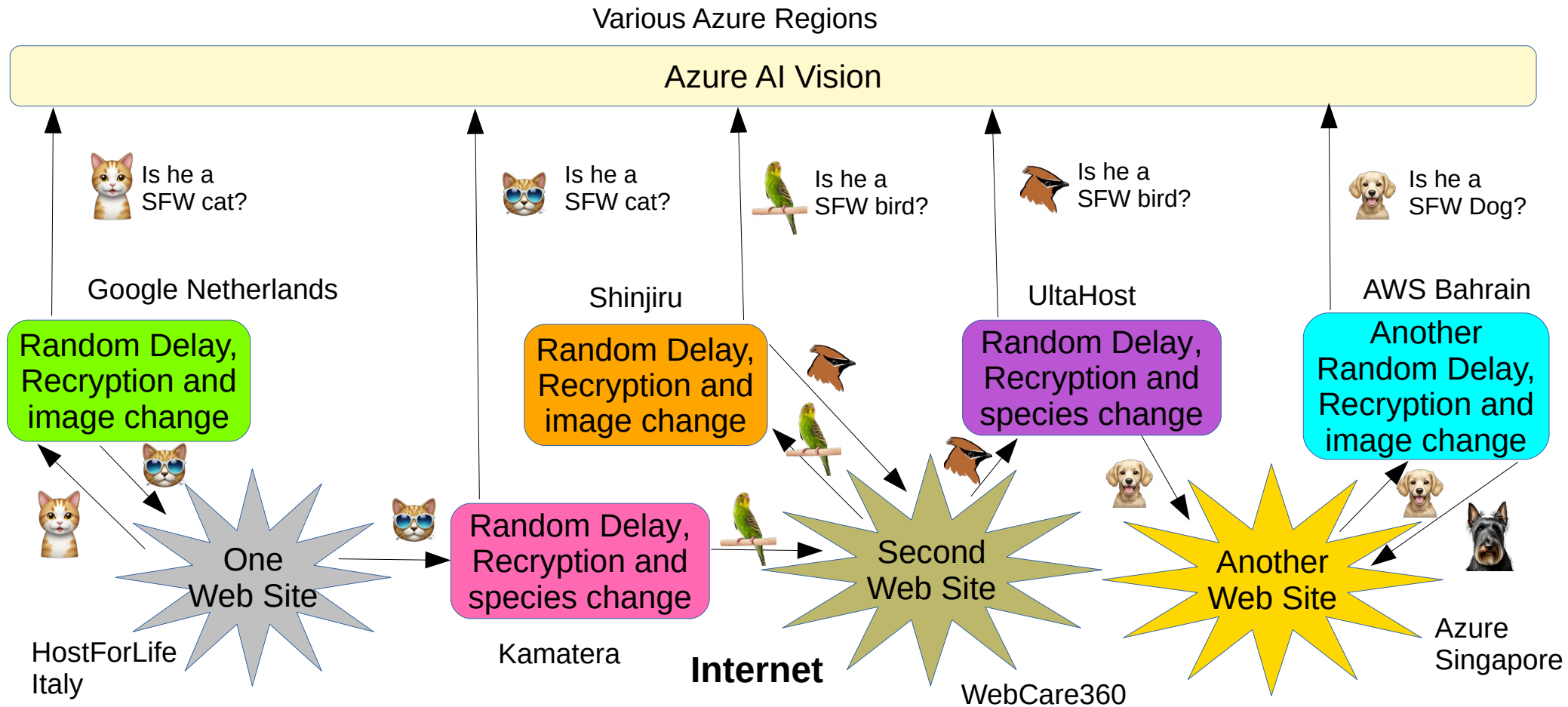
Cloudflare or On-Prem or in same data center as Client

Stable Diffusion with web services wrapper running in Docker Containers



For operational security images should never be re-used

AI Safety Layer



Benefits

1. Supports groups who “do not want to know each other”
2. Does not draw attention to groups who are conversing
3. Let groups control their own encryption and keys locally

Issues

1. Wasteful of Resources

- All messages need to be downloaded for inspection
- Recipient has to make multiple attempts to decrypt
 - Messages meant for them (and succeed)
 - Messages meant for others (and fail)

2. Due to the delay, real-time voice or video not supported

3. Using SignalR is expedient but limits message size

Use Cases

“Good”

- ✓ Reporter and source
- ✓ Politician and donor
- ✓ Agent and handler

“Bad”

- x Arms merchant and customer
- x Co-conspirators
- x Nudes

POC Goals

1. Stable Diffusion with web services wrapper running in an on-prem Docker Container
2. Two instances of SignalR web sites similar to Azure SignalR running on-prem
3. Two instances of unidirectional species-changing Proxies to transfer messages between them running on-prem
4. An instance of an image-changing Helper to support each web site (two total) running on-prem
5. Two instances of console-based Client attached to each web site (4 total) running on-prem supporting two 3-party conversations

Could this be a Phone App?

- Yes, but it weakens operational security because the hidden message would be encrypted and decrypted on a device (the phone) which is connected to the internet.
- The following slides mock up of that solution
- It would be a really bad idea to run the client in a browser using code served from the internet

Possible Client Setup Wireframe

Mode ▾	Select ▾
>Keys	>Server
>Exit	>Account
	>Add New

Welcome to Messaging in Plain Sight

Please enter your server key, your account key and add conversation key(s) to begin

Mode ▾	Select ▾
>Keys	>Server
>Exit	>Account
	>Add New

Key: Server
Passphrase: Three people can share a secret if one is dead
Date: Aug 21, 1932
Time: 9:30:30 PM

Server Information
Animal: Cat
IP: 79.120.10.74

SAVE

Mode ▾	Select ▾
>Keys	>Server
>Wipe	>Account
>Exit	>Add New

Key: Account
Passphrase: We are the champions
Date: Nov 22, 1963
Time: 12:30:00 PM

SAVE

<= There are two yellow and orange drop down (select) controls at the top of each page which are always shown as "dropped down"

Wipe will remove server and account information

Possible Convo Setup Wireframe

Mode ▾	Select ▾
>Keys	>Server
>Wipe	>Account
>Exit	>Add New

Key: DevOps
Passphrase: Start drinking heavily
Date: Oct 14, 1066
Time: 05:49:33 PM

SAVE

Mode ▾	Select ▾
>Keys	>Server
>Convos	>Account
>Wipe	>DevOps
>Exit	>Add New

Key: Boss
Passphrase: Death before downtime
Date: Jan 14, 1966
Time: 09:00:01 AM

SAVE

Mode ▾	Select ▾
>Keys	>Server
>Convos	>Account
>Wipe	>DevOps
>Exit	>Boss
	>Add New

Key: Fam
Passphrase: I am wasted and cannot find my way home
Date: Jan 1, 1997
Time: 00:00:01 AM

SAVE

Wipe will remove all keys and the **local** copy of any associated conversations

Possible Client Use Wireframe

Mode ▾

>Keys
>**Convos**
>Clear
>Wipe
>Exit

Select ▾

>**DevOps**
>Boss
>Fam

Sep 6, 2024 9:29:11 AM
Hey, its Joe. The
Exchange Server is
down.

Sep 6, 2024 9:30:23 AM
They are all down,
freaking power is
out again and the
generator croaked

>

SEND

Mode ▾

>Keys
>**Convos**
>Clear
>Wipe
>Exit

Select ▾

>**DevOps**
>Boss
>Fam

ower is
and the
generator croaked

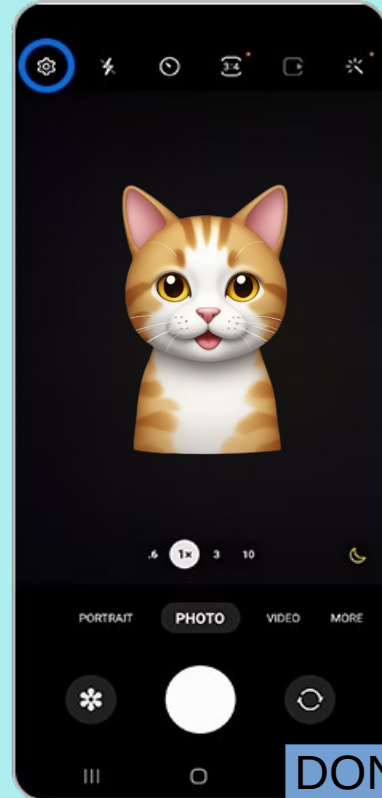
Sep 6, 2024 9:31:30 AM



> LOL

SEND

Please take a picture of
a Cat.



DONE

Mode ▾

>Keys
>**Convos**
>Clear
>Wipe
>Exit

Select ▾

>**DevOps**
>Boss
>Fam

croaked

Sep 6, 2024 9:31:30 AM



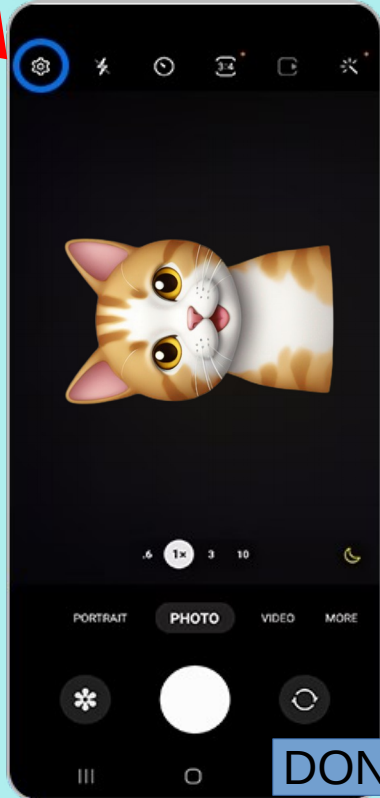
Sep 6, 2024 9:32:47 AM
LOL

>

SEND

Unlike original design, this client is trusted with the plaintext of the message for convenience

Possible Client Abuse Wireframe

Mode ▾	Select ▾		Mode ▾	Select ▾		Mode ▾	Select ▾
>Keys	>DevOps	Please take a picture of a Cat.  > It's Jim. I feel sick and need to take a sick day. Joe is on. SEND	>Keys	>DevOps	Sep 6, 2024 9:34:15 AM It's Jim. I feel sick and need to take a sick day. Joe is on. > SEND	>Keys	>DevOps
>Convos	>Boss		>Convos	>Boss		>Convos	>Boss
>Clear	>Fam		>Clear	>Fam		>Clear	>Fam
>Wipe			>Wipe			>Wipe	
>Exit			>Exit			>Exit	
						Sep 6, 2024 9:31:30 AM	
							
						Sep 6, 2024 9:32:47 AM	
						LOL	CLEAR

Clear will remove the **local** copy of some or all conversation contents